

From signatures to populations: when does single-cell information improve drug retrieval?

Abstract

Artificial intelligence virtual cells (AIVCs) aim not only to predict how cells respond to perturbations, but ultimately to help prioritize which interventions to test. Transcriptomic drug retrieval has traditionally represented each perturbation by a single expression signature, whereas single-cell perturbation atlases now make it possible to compare entire response populations. Here we formulate population-to-population drug retrieval and ask when this additional single-cell information improves the ranking of candidate drugs. Within the score family studied here, classical mean-signature retrieval is the zero-variance limit of population retrieval, directly connecting the two regimes. Population-level retrieval captures substantial signal beyond the mean, but capturing more information does not necessarily produce a better drug ranking. When the same rankings are judged by criteria less tied to the retrieval score, large gains under response matching disappear or reverse under mechanism-of-action recovery, and most of the advantage that remains against an external functional readout can be reproduced by response magnitude alone. More strikingly, two external protein-based evaluators built from the same cells favour opposite methods when only the statistical form of the evaluator is changed. Together, these results establish population-to-population retrieval as a distinct regime for single-cell drug search and show that its value depends on whether heterogeneous responses are present, recoverable and consequential enough to change candidate choice.

Keywords: artificial intelligence virtual cell; single-cell perturbation; drug retrieval; population-to-population retrieval; cellular heterogeneity; objective–utility mismatch

Introduction

Artificial intelligence virtual cells (AIVCs)^{1,2} are being developed not only to predict how cells respond to perturbations^{3–5}, but ultimately to help decide which perturbations are worth testing. These goals define two complementary problems. Forward prediction asks what response a perturbation will produce^{6–10}; retrieval asks which perturbation best matches a cellular response of interest. Transcriptomic drug retrieval has traditionally addressed the latter by representing each drug with a single perturbation signature and comparing these signatures by similarity or reversal^{11–13}. This representation has made large drug libraries searchable, but it compresses the diversity of responses measured by single-cell experiments into one summary vector. Single-cell perturbation atlases now make it possible to retain that diversity^{14–16}, including differences between cell states¹⁷, minority responses¹⁸ and multiple response patterns within the same population. Drug retrieval can therefore move from comparing mean signatures to comparing entire response populations. We refer to this as population-to-population drug retrieval and ask when this additional single-cell information actually improves the ranking of candidate drugs.

39 More detailed information, however, is not automatically more useful information. Two drugs can
40 produce similar average responses while affecting different subsets of cells, and a population-aware
41 method can distinguish them even when a mean-based method cannot. But this extra discrimination
42 matters only if it leads to a better drug ranking under a biologically relevant criterion. This creates
43 a fundamental evaluation problem^{19–21}. If a distribution-aware method^{22–24} is judged mainly by how
44 well it matches full response distributions^{25,26}, then it is being evaluated on the same type of information
45 it was designed to preserve; a large gain may therefore show that the method captures distributional
46 structure well without showing that it makes a better drug choice^{27,28}. The reverse can also happen if
47 the evaluator reduces the response to a mean, thereby favouring mean-based retrieval²⁹. We refer to this
48 gap between capturing additional information and demonstrating its value for the final ranking as an
49 objective–utility mismatch. The key question is therefore not whether population-level retrieval can use
50 more single-cell information, but when that information is genuinely relevant to the retrieval decision.

51 To answer this question, the information used for retrieval must be separated from the criterion
52 used to judge the result. We therefore formulate drug retrieval directly between single-cell response
53 populations and use a controlled design in which the query, candidate library and ranked list remain
54 unchanged while only the evaluation criterion is varied. Mean-signature retrieval is not a separate task
55 in this framework: within the score family considered here, it is the zero-variance limit of population
56 retrieval, linking classical signature matching to full population comparison. We then judge the same
57 rankings at increasing distance from the retrieval objective, from response-matching criteria, to biolog-
58 ical criteria such as mechanism recovery, and finally to external functional readouts^{30,31}. This design
59 allows us to ask directly whether retaining population structure merely improves how well a response
60 is represented, or whether it improves the ranking of candidate drugs.

61 The two are not equivalent. Population-level retrieval captures substantial signal beyond mean sig-
62 natures, showing that single-cell response populations contain retrievable information that averaging
63 removes. But much of this apparent advantage weakens when the same rankings are judged by criteria
64 that are less tied to the retrieval score: gains under response-matching evaluation disappear or reverse
65 under mechanism recovery, and most of the advantage that remains against an external functional read-
66 out can be reproduced by a simple measure of response magnitude that uses no distributional informa-
67 tion. More strikingly, two external protein-based evaluators built from the same cells favour opposite
68 methods when only the statistical form of the evaluator is changed. External evaluation, in other words,
69 is not automatically neutral. Together, these results suggest a practical framework for when population-
70 level retrieval can add value beyond a mean signature: different cell states must respond differently,
71 those differences must be recoverable from the observed population, and the additional information
72 must ultimately be important enough to change which candidate is preferred. Population-to-population
73 retrieval therefore defines a distinct regime for single-cell drug search, but its value depends not simply
74 on preserving more cellular detail, but on whether that detail changes the decision.

75 Results

76 Population-to-population retrieval generalizes classical signature matching

77 We first asked what changes when the object used for retrieval is a population of single cells rather than
78 a collapsed expression signature. For a query population Q and a library of candidate perturbation pop-

ulations $\{P_d\}$, population-to-population retrieval ranks candidates according to the similarity of their cellular responses (Fig. 1). Classical mean-signature retrieval^{11,12} performs the same task after replacing each population with a single differential-expression vector. Population-level retrieval instead retains the individual cells and can therefore use response structure that is lost on averaging, including differences between cellular states or subpopulations. The retrieval problem is unchanged; what changes is how much of the measured response is retained before candidate ranking.

This distinction also changes what must be shown for the additional information to be useful. Two candidate populations can have the same mean response but differ markedly in how individual cells respond (Fig. 1a–c). A population-aware score can distinguish such candidates, whereas a mean-based score cannot. However, a gain measured by a criterion that rewards the same population structure establishes that the structure has been captured, not necessarily that the resulting ranking is more biologically useful. We therefore use the same candidate rankings throughout and judge them under criteria that are progressively less tied to the retrieval score: response matching, task-proximal biological criteria such as mechanism recovery, and externally measured functional readouts (Fig. 1d,e). This controlled evaluation separates the information used to construct a ranking from the criterion used to judge it.

Mean-signature and population-level retrieval are not separate tasks. Within the score family considered here, they are connected by the amount of within-population variation retained. Writing a cellular response as

$$X = \mu + \lambda\epsilon, \quad E[\epsilon] = 0,$$

the parameter λ controls the variation around the population mean. As λ approaches zero, the population collapses onto μ and the corresponding population comparison reduces to its mean-only endpoint (Fig. 2a–c). Mean-signature retrieval is therefore the zero-variance limit of population retrieval within this framework.

This connection is exact for the implemented mean-cosine baseline. Mean-cosine retrieval and the implemented CMap-style cosine score apply the same operation to the same mean differential-expression vectors and consequently give identical Hit@1 (0.388492; Fig. 2d). The canonical L1000 weighted connectivity score (WTCS) is related but not identical: it uses rank-based enrichment and yields a different Hit@1 (0.4635). Thus, the equivalence concerns the cosine connectivity rule used here rather than every signature-based retrieval score.

The same continuum extends beyond the zero-variance limit. In the subpopulation-coverage score, a temperature parameter β controls how strongly poorly matched subpopulations influence the final distance. On the controlled benchmark, increasing β moves the score continuously from mean-like aggregation (0.7125 as $\beta \rightarrow 0$) towards the worst-matched subpopulation (1.300 as $\beta \rightarrow \infty$; Fig. 2e). The global energy distance is recovered as the single-partition ($K = 1$) case of the same construction. These relationships place classical signature matching and population comparison on a common axis of sensitivity to within-population structure rather than treating them as unrelated method classes (Fig. 2f).

111 Cell populations contain retrievable information beyond mean signatures

112 Having established the connection between mean-signature and population-level retrieval, we next
 113 asked whether retaining within-population structure provides additional information for retrieval in
 114 practice. We first considered the setting in which candidate responses are directly observed, so that the

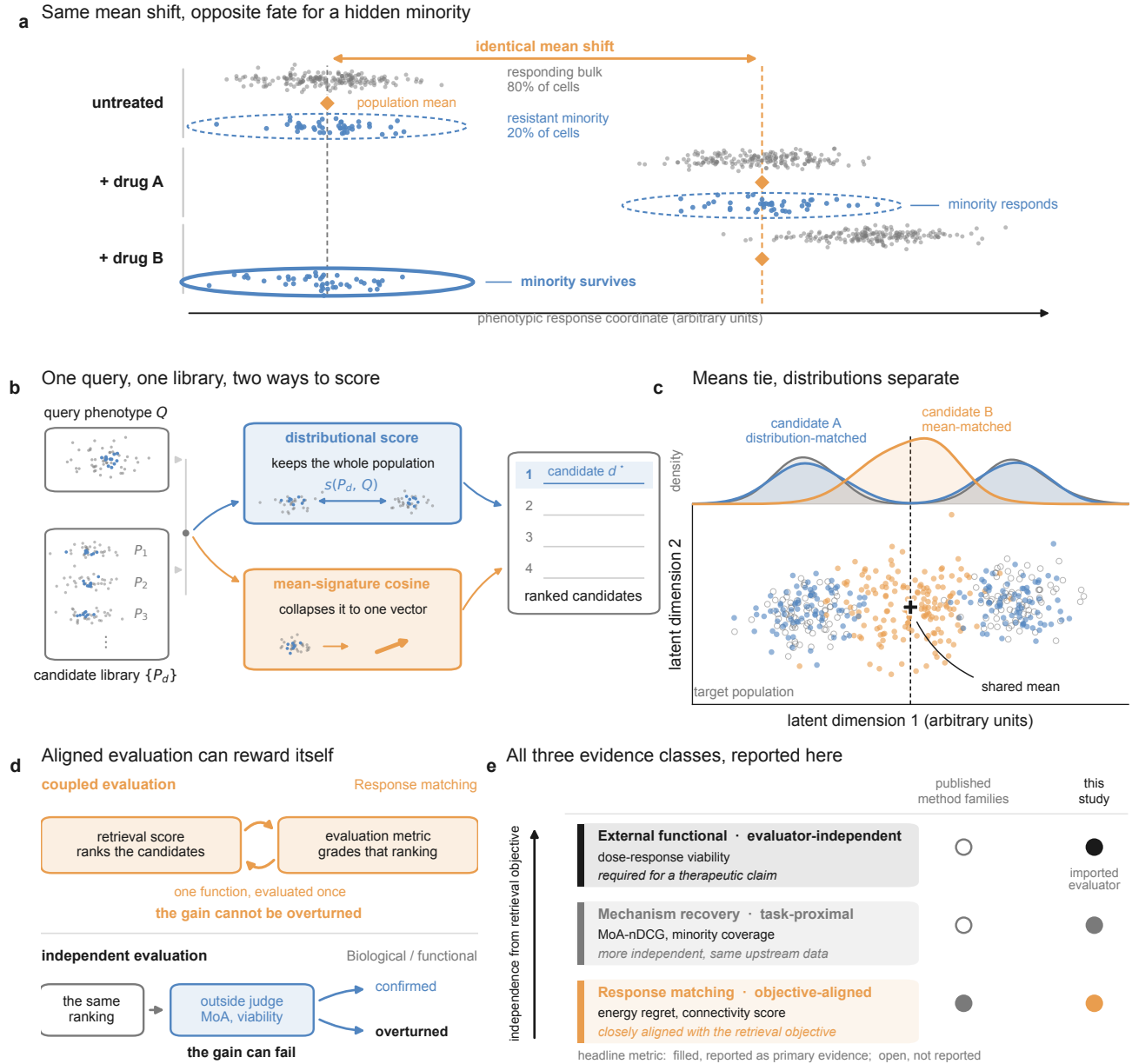


Figure 1: From perturbation signatures to population-to-population drug retrieval. **a**, Two drugs can produce the same mean transcriptional response while acting differently across cells. In the schematic example, one sub-population remains largely unchanged under drug B while another responds more strongly, preserving the same population mean. **b**, The retrieval task. A query response is matched against a library of candidate responses. Mean-signature retrieval reduces each response population to a single differential-expression vector, whereas population-level retrieval compares the cellular populations directly; both return a ranked candidate list. **c**, Candidate populations with the same mean can remain distinguishable through their within-population structure, illustrated here by unimodal and bimodal responses. **d**, The criterion used to evaluate a ranking can remain closely aligned with the information used to construct it or can be moved progressively further from the retrieval objective. Objective-aligned evaluation tests fidelity to response matching, whereas more independent biological or functional criteria can confirm or overturn the resulting ranking. **e**, Evaluation framework used in this study: response-matching evaluation, task-proximal biological evaluation such as mechanism recovery, and external functional evaluation. Representative method families are shown according to the evidence used in their headline evaluation; the analyses here evaluate fixed rankings across all three levels. The external functional drug-response analysis uses a semi-real readout measured in cell lines not used to construct the transcriptional rankings.

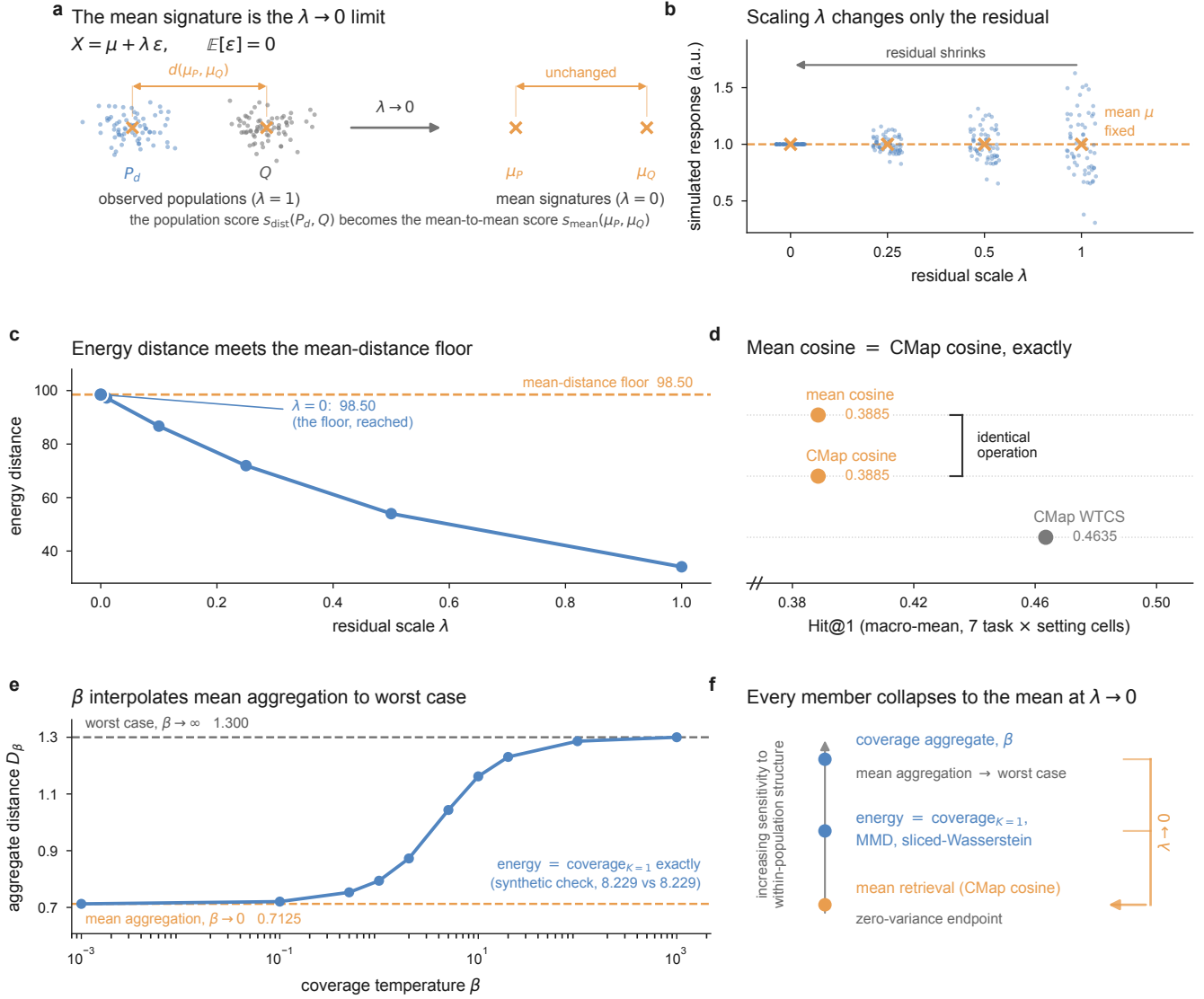


Figure 2: Mean-signature retrieval is the zero-variance limit of population retrieval. **a**, A cellular response is written as $X = \mu + \lambda \epsilon$, where μ is the population mean and λ controls within-population variation. **b**, Changing λ alters the spread of the response population while leaving its mean unchanged. **c**, As within-population variation vanishes, population-level distance converges to its mean-only endpoint. **d**, Under the cosine rule implemented here, mean-cosine retrieval and CMap-style cosine retrieval are numerically identical (Hit@1, 0.388), whereas the rank-based WTCS score is related but not algebraically identical. **e**, The subpopulation-coverage temperature β provides a continuous interpolation from mean-like aggregation to emphasis on the worst-matched subpopulation; energy distance corresponds to the single-partition case. **f**, Classical mean-signature retrieval and population-level retrieval can therefore be viewed within one family that differs in sensitivity to within-population structure.

115 full cellular populations are available to both retrieval approaches. Across the seven task-setting com-
116 binations analysed here, energy-distance retrieval reached a Hit@1 of 0.837, compared with 0.388 for
117 mean-cosine retrieval and the numerically identical CMap-style cosine baseline^{11,12} (Fig. 3a). The same
118 pattern was observed in PCA space, where population distance reached 0.778 compared with 0.518 for
119 the corresponding mean-based score. These values are unweighted macro-means across task-setting
120 combinations; weighting by the number of queries gave 0.887 for energy and 0.421 for mean cosine
121 (Extended Data Fig. 1 and Supplementary Table 2).

122 The advantage was substantial on average but not universal. In Frangieh³², the natural biological
123 dataset contributing to this comparison, mean-signature retrieval slightly outperformed energy distance
124 (Hit@1 0.600 versus 0.578; Fig. 3b). Thus, retaining the full response population can reveal information
125 that is lost by averaging, but whether that information improves retrieval depends on the task.

126 We next asked whether this advantage persists when only part of each candidate population is ob-
127 served. In this controlled partial-observation setting, candidates were evaluated using a distribution-
128 sensitive regret criterion that penalizes selecting a response population with poorer latent per-cell out-
129 comes than the best available candidate. Across 765 queries, the worst-case subpopulation-coverage
130 score reduced median regret by +0.118 relative to mean retrieval (mean +0.258), with lower regret on
131 72% of queries (paired Wilcoxon $p = 1.6 \times 10^{-66}$; Fig. 3c). The corresponding median was +0.119 among
132 the 621 queries identified by the pre-specified diagnostic as favourable to population-level retrieval.

133 The signal was not confined to a single distributional score: all five population-level variants showed
134 positive gains under the same response-matching criterion (Fig. 3f). At the same time, simple diagnostics
135 did not reliably identify where those gains would occur. Queries flagged as favourable and unfavourable
136 showed nearly identical median regret reductions (+0.119 and +0.122, respectively; Fig. 3d), and pro-
137 gressively merging the two constructed subpopulations did not produce a consistent monotonic loss of
138 the energy-distance advantage across cell lines (Fig. 3e).

139 These results establish the first part of the problem: single-cell response populations contain retriev-
140 able structure beyond their mean signatures. They do not establish that this additional structure im-
141 proves a biologically meaningful drug ranking. Both source identification and the partial-observation
142 regret criterion reward correspondence between response populations, and therefore remain closely
143 aligned with the information used by population-level retrieval. We next asked whether the same ad-
144 vantage survives when the rankings are judged by biological and functional criteria that are less directly
145 tied to response matching.

146 **Additional retrieval information does not necessarily improve biological ranking**

147 The preceding analyses show that population-level retrieval can exploit response structure that is lost on
148 averaging. We next asked whether this additional information improves the same candidate rankings
149 when success is defined by biological criteria rather than by response matching itself. To isolate the effect
150 of the evaluator, we held the query, candidate library, retrieval scorer and resulting ranking fixed and
151 changed only the criterion used to judge that ranking (Fig. 4).

152 We first used mechanism-of-action recovery as a task-proximal biological criterion. Among the
153 600 partial-observation queries for which both response-matching and mechanism annotations were
154 available, the worst-case subpopulation-coverage scorer showed a median improvement of +0.129 over
155 mean retrieval under the response-matching criterion. When the identical rankings were evaluated by

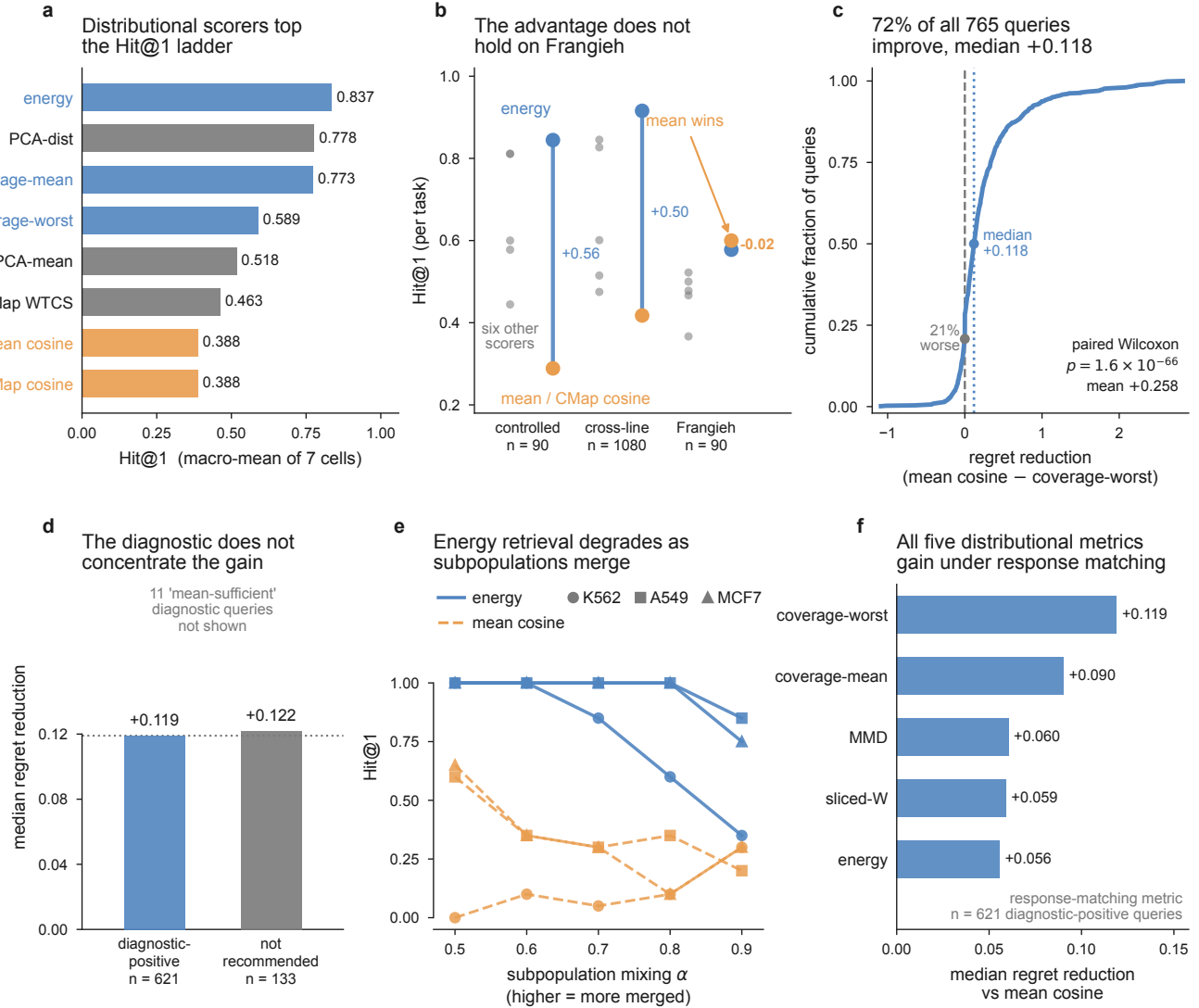


Figure 3: Cell populations contain retrievable information beyond mean signatures. **a**, Hit@1 across retrieval approaches, shown as unweighted macro-means over seven task-setting combinations. Energy-distance retrieval reaches 0.837 compared with 0.388 for mean cosine and the numerically identical CMap-style cosine baseline. **b**, Per-task Hit@1. Population-level retrieval performs better on the controlled and cross-line SciPlex3 tasks but not on Frangieh, where mean/CMap cosine reaches 0.600 compared with 0.578 for energy distance. **c**, In the partial-observation setting, the worst-case subpopulation-coverage score reduces regret relative to mean retrieval by a median of +0.118 across all 765 queries, with improvement on 72% of queries. **d**, A pre-specified diagnostic intended to identify queries favourable to population-level retrieval does not concentrate the gain: median regret reduction is +0.119 among 621 recommended queries and +0.122 among 133 non-recommended queries. **e**, Increasing the similarity of the two constructed subpopulations does not consistently eliminate the energy-distance advantage. In K562, the Hit@1 difference falls from 1.00 to 0.05 as α increases from 0.5 to 0.9, whereas A549 and MCF7 do not show the same monotonic pattern. **f**, All five population-level scoring variants show positive regret reduction under the response-matching criterion.

156 mechanism-of-action recovery, however, the median difference was 0.000 and the mean was -0.037
157 (Fig. 4a,b). The change was not driven by a small number of extreme queries: mechanism recovery
158 decreased for 41.3% of queries, increased for 34.2%, and was unchanged for 24.5% (paired Wilcoxon
159 $p = 2.4 \times 10^{-4}$). The same qualitative pattern was observed across all five population-level scoring vari-
160 ants. Thus, the large gain in response matching does not translate directly into improved recovery of
161 biologically related drugs.

162 A second task-proximal criterion asked whether population-level retrieval better preserved minority-
163 state responses. Across 239 real-data retrieval tasks, the mean improvement over mean retrieval was
164 only $+0.0018$, and in 102 tasks the two approaches selected the same candidate and therefore produced
165 exactly the same score (Fig. 4c,g). The small positive difference was concentrated in the constructed
166 cross-cell-line mixtures ($+0.0043$), whereas within-line SciPlex3, Frangieh and CD34+ showed gains
167 close to zero. Stratifying queries by their measured response divergence did not reveal a hidden regime
168 of large benefit: the minority-state advantage remained small even in the most divergent queries, and
169 mechanism-of-action recovery showed no significant positive gain in that stratum (Fig. 4c). These results
170 indicate that stronger population differences do not, by themselves, guarantee a more biologically useful
171 ranking (Extended Data Fig. 2 and Supplementary Note 2).

172 We next asked whether population-level retrieval carries information that is visible to an external
173 functional readout. For the SciPlex3 drug panel, we constructed a drug-drug functional similarity mea-
174 sure from GDSC2^{30,31} dose-response profiles across 966 cell lines, excluding the three cell lines used
175 to generate the transcriptional queries. After removing the shared cell-line sensitivity effect, energy-
176 distance retrieval reached a Spearman correlation of $+0.276$ with this external functional similarity, com-
177 pared with $+0.083$ for the corresponding mean-signature baseline (Fig. 4h). This provides an affirmative
178 result: population-level transcriptional similarity contains functional information that is not captured as
179 strongly by the mean signature (Extended Data Fig. 5 and Supplementary Table 4).

180 Much of that advantage, however, did not require comparing cellular distributions. Ranking candi-
181 dates only by how closely their overall response magnitude matched the query reached $+0.232$, despite
182 using a single scalar per perturbation. After accounting for this magnitude channel, the remaining as-
183 sociation for energy retrieval was $+0.097$, positive in all three cell lines but substantially smaller than
184 its unadjusted value (Fig. 4h,i). Energy outperformed the magnitude-only ranking in 62 of 103 queries,
185 but this difference was not significant under the nominal paired comparison ($p = 0.073$), which is itself
186 anticonservative because queries within a cell line share the same candidate pool. Population-level re-
187 trieval therefore carries external functional signal, but much of the signal detected here can be recovered
188 without using cell-to-cell distributional structure.

189 The conclusion is consequently more specific than either a positive or a negative verdict on
190 population-level retrieval. Retaining single-cell populations clearly increases the information available
191 for response matching, and some of that information extends to an external functional readout. But
192 the size—and sometimes even the direction—of the apparent advantage depends on what aspect of
193 the response is used to judge the ranking. This motivates a further question: does using an external
194 readout remove this dependence, or can the statistical form of an external evaluator itself determine
195 which retrieval strategy appears superior?

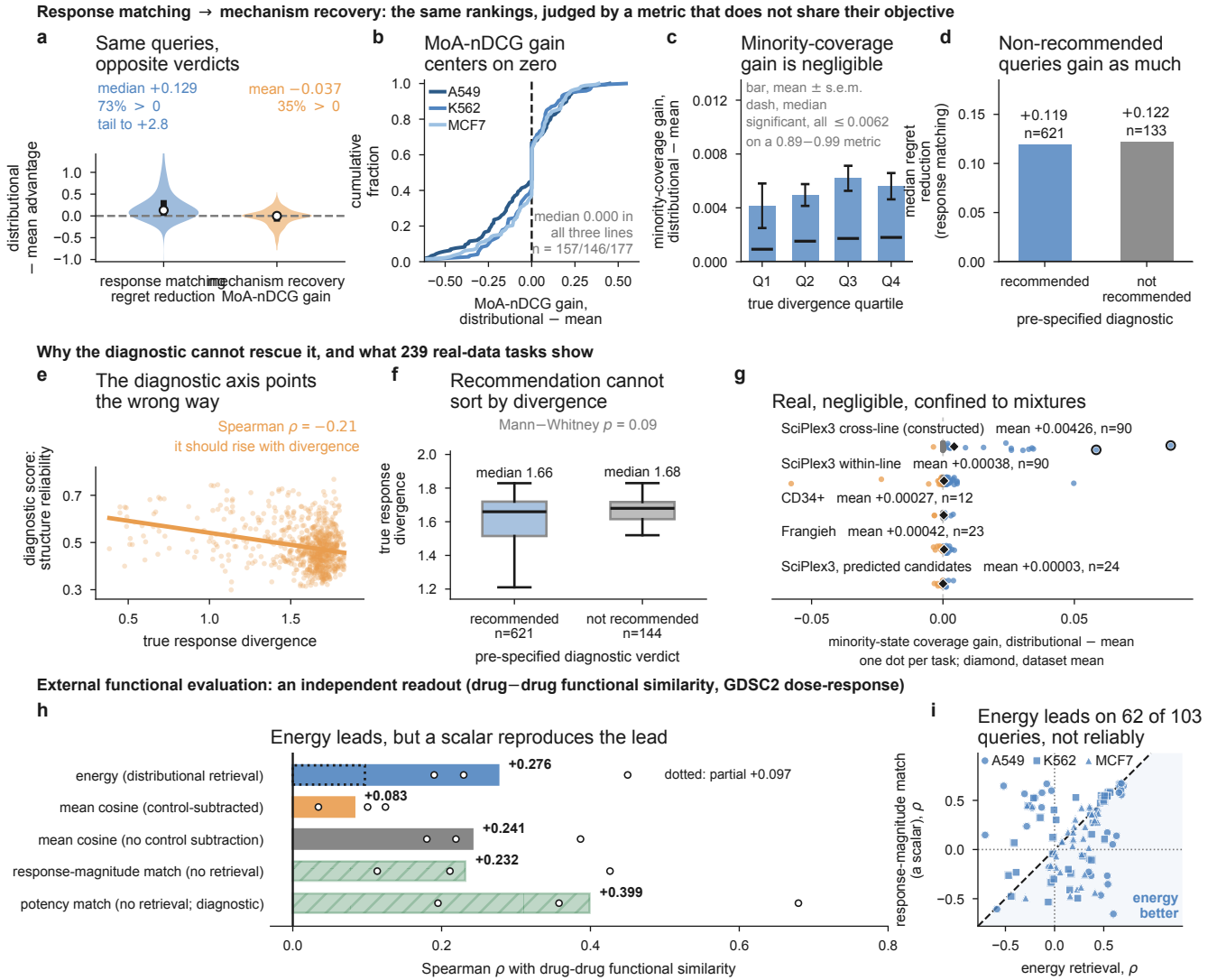


Figure 4: Additional population information does not necessarily improve biological ranking. **a**, The same population-level scorer and the same 600 queries are evaluated under two criteria. A median advantage of +0.129 over mean retrieval under response matching becomes 0.000 under mechanism-of-action recovery (mean difference, -0.037). **b**, Mechanism-of-action recovery by cell line, showing no consistent advantage for population-level retrieval. **c**, Minority-state coverage as a function of measured response divergence. The positive difference remains small across divergence strata, and increased divergence does not produce a corresponding gain in mechanism recovery. **d**, A pre-specified diagnostic does not identify the queries on which population-level retrieval gains most: median response-matching gains are similar in the recommended and non-recommended groups. **e,f**, Diagnostics of the same rule show that its internal axes are weakly or inversely related to measured response divergence, explaining why it does not concentrate the observed gain. **g**, Across 239 real-data tasks, the improvement in minority-state coverage is small overall (mean +0.0018) and is concentrated in constructed cross-cell-line mixtures; 102 tasks yield exactly the same candidate under both retrieval approaches. **h**, External functional evaluation using GDSC2 drug–drug response similarity. Energy retrieval reaches $\rho = +0.276$ compared with +0.083 for mean-signature retrieval. A scalar that matches only response magnitude reaches +0.232, and accounting for this channel reduces the energy association to +0.097. **i**, Per-query comparison between energy retrieval and the response-magnitude baseline. Energy performs better on 62 of 103 queries; the nominal paired $p = 0.073$ is anti-conservative because queries within a cell line share a candidate library.

External evaluation is not automatically neutral

The preceding results show that the apparent value of population-level information changes when a ranking is judged by a criterion different from the one used to construct it. A natural response is therefore to evaluate retrieval against an external measurement that the retrieval algorithm never observes. We asked whether such an external readout is sufficient to provide a neutral judge.

Frangieh Perturb-CITE-seq³² provides a direct test because RNA and surface proteins are measured from the same cells, while the retrieval algorithms use only the RNA. We constructed two protein-based evaluators from the same cells and the same protein measurements, differing only in how those measurements were summarized (Fig. 5c). The first compared the mean protein responses of each query–candidate pair. Under this mean-based evaluator, mean-signature retrieval outperformed energy-distance retrieval (+0.242 versus +0.146), consistently across Control, IFN γ and Co-culture conditions.

We then evaluated the identical RNA-derived rankings against the full distributions of the same protein responses rather than their means. The ordering reversed. Energy-distance retrieval reached +0.529, compared with +0.334 for mean-signature retrieval, with the same direction in all three immune conditions (Fig. 5c). Neither retrieval method had access to either protein-based evaluator. The reversal therefore does not arise from a method being scored directly by its own training objective. Instead, it shows that even an external and unseen readout can favour a retrieval method whose statistical representation matches the form in which that readout is evaluated.

A response-magnitude control refined this result. A scalar score that compares only the overall magnitude of the query and candidate responses rose from +0.125 under the mean-based protein evaluator to +0.352 under the distribution-based evaluator, showing that part of the reversal can occur without comparing cellular distributions at all. Nevertheless, energy distance exceeded this scalar by +0.177 under the distribution-based evaluator, compared with only +0.022 under the mean-based evaluator. The protein measurements therefore contain genuine population-level signal, but whether that signal appears advantageous depends strongly on which aspect of the external measurement is preserved by the evaluator.

A controlled benchmark illustrates the same dependence under known ground truth (Fig. 5a,b). In HIR-Bench, an analytical boundary separates settings in which subpopulation structure can alter the preferred candidate from those in which the mean is sufficient. However, this synthetic boundary does not transfer reliably to the real datasets analysed here. We therefore use HIR-Bench as a conceptual control for when population structure can matter, rather than as a diagnostic for deciding when it will matter in biological data (Extended Data Fig. 6 and Supplementary Note 1).

We next asked whether naturally heterogeneous tissue¹⁷ contains the biological response structure that population-level retrieval is intended to exploit. In glioblastoma samples from ZhaoSims2021, drug responses differed across co-existing cellular compartments: the median cosine between compartment-specific response directions was 0.566 across 17 patient–drug pairs, substantially lower than perfect alignment and therefore consistent with state-dependent drug responses (Fig. 5e). This heterogeneity was also recoverable in principle. For within-patient drug-response partitions, a supervised probe reached an accuracy of 0.923, whereas the best unsupervised procedure reached 0.777, leaving a median gap of +0.117 (Fig. 5d). In real tissue, the relevant information is therefore present, but current unsupervised recovery does not fully extract it (Extended Data Fig. 4 and Supplementary Note 4).

237 The remaining question is whether this additional structure is large enough to change which drug
 238 should be preferred. Here the tumour data provide a constraint rather than a retrieval result. Across 18
 239 within-patient drug pairs, similarity between mean drug-response signatures strongly tracked similarity
 240 between malignant-cell responses. Because these quantities partly share cells, we repeated the analysis
 241 using disjoint compartments: myeloid-response similarity still ranked malignant-response similarity at
 242 Spearman $\rho = +0.835$ (Fig. 5f). Thus, drugs that look similar at a broader population level also tend to
 243 look similar within a specific cellular compartment, leaving less room for within-population structure to
 244 reorder candidates than the existence of heterogeneity alone would suggest. No retrieval was performed
 245 on this patient cohort, so these analyses do not demonstrate the presence or absence of a population-
 246 level retrieval advantage. Instead, they identify the missing requirement that the next section addresses:
 247 additional cellular structure must not only exist and be recoverable, but must also be relevant enough to
 248 change candidate preference.

249 **When can population-level information improve drug retrieval?**

250 The preceding results suggest that retaining single-cell populations is useful only when the information
 251 they preserve can propagate all the way to candidate ranking. We therefore separated this problem
 252 into two measurable information requirements and a third, downstream requirement. First, different
 253 cellular states must respond differently to the same perturbation. Second, those differences must be
 254 recoverable from the observed response population. Even when both conditions hold, however, the
 255 additional information can improve retrieval only if it is large enough to change which candidate is
 256 preferred (Fig. 6a). We refer to these requirements as differential response, recoverability and decision
 257 relevance.

258 We first examined differential response in the candidate populations supplied to the retriever. When
 259 candidate responses were generated by perturbation predictors rather than directly observed, the ad-
 260 vantage of population-level retrieval became small and inconsistent: the change in nDCG@10 relative
 261 to mean retrieval was -0.016 for an average-effect predictor, -0.071 for nearest-neighbour transfer and
 262 $+0.027$ for a linear-latent predictor (Fig. 6b). The limiting case is simple. Any predictor that adds the
 263 same perturbation effect to every cell in a context gives every cellular state the same response vector
 264 and therefore cannot generate state-specific differential response. This is an algebraic property of the
 265 model class rather than a consequence of insufficient training (Extended Data Fig. 3 and Supplementary
 266 Note 3).

267 A within-context analysis confirmed that this distinction is biologically relevant. Real treated cells,
 268 partitioned by their pre-perturbation state, produced markedly different response directions, with a
 269 mean induced-response cosine of 0.205 across 30 cell-line and drug combinations. By contrast, the ad-
 270 ditive average-effect and linear-latent predictors returned a cosine of 1.000 by construction, while a de-
 271 ployed scGen model⁶ remained close to this additive limit at 0.972. A non-parametric optimal-transport
 272 map^{22,33} moved further away from the additive limit but remained substantially more aligned than the
 273 observed responses (0.900, s.d. 0.107; Supplementary Note 3). Thus, preserving the heterogeneity of the
 274 starting population is not sufficient: a predictor must also reproduce how the perturbation acts differ-
 275 ently across cellular states. We did not identify a predictor in these experiments that both transferred
 276 accurately across contexts and reproduced this differential response, so whether opening this informa-
 277 tion channel is sufficient to improve cross-context retrieval remains unresolved.

From the inside: analytically, inside a synthetic benchmark, and between two real evaluators

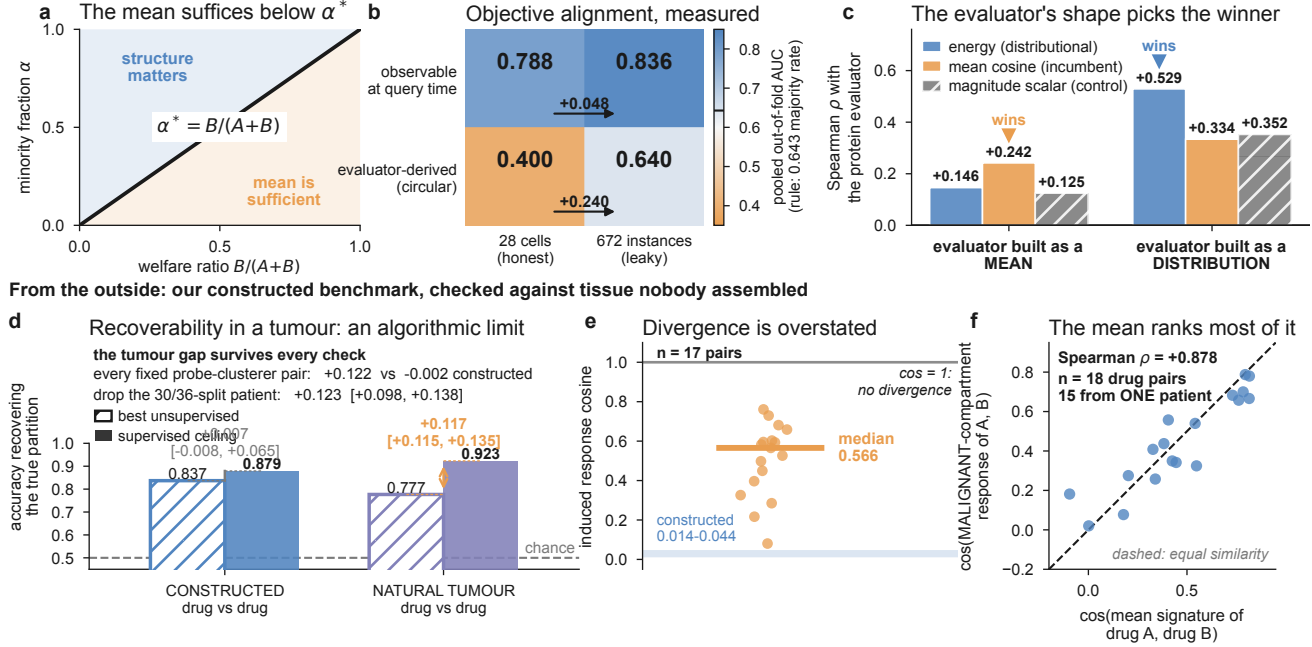


Figure 5: External evaluation remains sensitive to statistical form, while natural tissue constrains the value of population structure. **a**, HIR-Bench provides a controlled setting in which the analytical boundary $\alpha^* = B/(A+B)$ separates regimes where the mean is sufficient from those where subpopulation structure can change the preferred candidate. **b**, Retrieval failure is moderately predictable from information observable at query time (AUC 0.788, majority-class rate 0.643), whereas features derived from the benchmark's latent utility construction do not generalize beyond chance. The synthetic boundary is therefore used as a conceptual control rather than a diagnostic for real data. **c**, Two external protein-based evaluators are constructed from the same cells and protein measurements but summarize them differently. Under the mean-based protein evaluator, mean-signature retrieval outperforms energy distance (+0.242 versus +0.146); under the distribution-based evaluator, the ordering reverses (+0.334 versus +0.529). The reversal holds in all three immune conditions. A response-magnitude scalar increases from +0.125 to +0.352 as the evaluator becomes distributional, but energy retains a +0.177 advantage over this scalar under the distribution-based evaluator. **d**, Recoverability of within-patient drug-response structure in glioblastoma tissue. A supervised probe reaches 0.923 compared with 0.777 for the best unsupervised method, a median gap of +0.117, indicating that response information is present but not fully recovered by unsupervised partitioning. **e**, Different cellular compartments in the same tumour respond differently to the same drug: the median cosine between compartment-specific response directions is 0.566 across 17 patient-drug pairs. **f**, Mean-level drug similarity nevertheless tracks compartment-specific response similarity. Using disjoint cellular compartments, myeloid-response similarity ranks malignant-response similarity at Spearman $\rho = +0.835$. No retrieval was performed on this cohort; the analysis tests the mechanism through which population structure could alter candidate preference rather than measuring a retrieval advantage directly.

278 Differential response is also not sufficient if the relevant structure cannot be recovered from the ob-
279 served population. We tested recoverability in a controlled two-state mixture with known labels. The
280 best unsupervised method reached an accuracy of 0.674, but the supervised ceiling was only 0.692, indi-
281 cating that little additional recoverable information remained in this constructed representation (Fig. 6e).
282 This apparent ceiling was partly a consequence of pooling multiple drugs into broad classes: posing the
283 same problem at the level of individual drug pairs raised the supervised ceiling to 0.879. Natural tissue
284 showed a different limitation. As shown in Fig. 5d, glioblastoma responses were highly decodable under
285 supervision (0.923) but less well recovered by unsupervised methods (0.777), indicating that recoverabil-
286 ity can become an algorithmic rather than an informational bottleneck.

287 The final requirement is that the recovered information must alter the candidate ordering itself. This
288 condition is not directly measured as the first two are, but several analyses constrain it. Stratifying the
289 retrieval queries by their measured response divergence did not reveal a regime in which stronger state-
290 specific responses produced a clear mechanism-of-action advantage: even the most divergent quartile
291 showed no significant positive gain (mean $+0.003$, $q = 0.57$), whereas the lowest-divergence quartile was
292 significantly negative (Fig. 6g). Similarly, the tumour analysis showed that compartment-specific drug
293 responses can differ while preserving much of the ordering already present at the broader population
294 level (Fig. 5f). Thus, heterogeneity can be biologically real and recoverable without being large enough
295 to reorder candidate drugs.

296 The same pattern was reproduced at larger scale in Tahoe-100M¹⁶. Across 3,630 drug-cell-line con-
297 ditions, the median cosine between state-specific response directions was 0.739, with fewer than 0.11%
298 of conditions as divergent as the constructed mixtures used earlier. The recoverability gap nevertheless
299 remained measurable across 960 drug pairs and 48 cell lines (median gap, 0.157), while a disjoint com-
300 parison of state-specific response ordering remained high ($\rho = 0.841$ across 44 evaluable contexts). These
301 measurements reinforce the distinction between the three requirements: differential responses can exist,
302 and can be recoverable, while still leaving limited room for population structure to change candidate
303 preference (Extended Data Fig. 7 and Supplementary Note 4).

304 Taken together, these results define when population-to-population retrieval can plausibly add value
305 beyond a mean signature. The candidate responses must contain state-specific perturbation effects; those
306 effects must be recoverable from the measured population; and, ultimately, they must carry informa-
307 tion that changes the ranking of candidate drugs. The first two requirements determine whether useful
308 population-level information is available. The third determines whether that information matters for the
309 retrieval decision.

310 Discussion

311 Single-cell perturbation atlases¹⁴⁻¹⁶ change the representation available for drug retrieval: a perturbation
312 can now be described by a population of cellular responses rather than by a single averaged signature.
313 Our results show that this added resolution matters, but not simply because preserving more cellular
314 detail produces a better ranking. Population-level retrieval recovers information that is lost by averag-
315 ing, yet whether that information improves a ranking depends on how success is defined. Large gains
316 under response matching can disappear under mechanism-of-action evaluation, and even external read-
317 outs can favour opposite retrieval strategies depending on whether they are summarized as means or

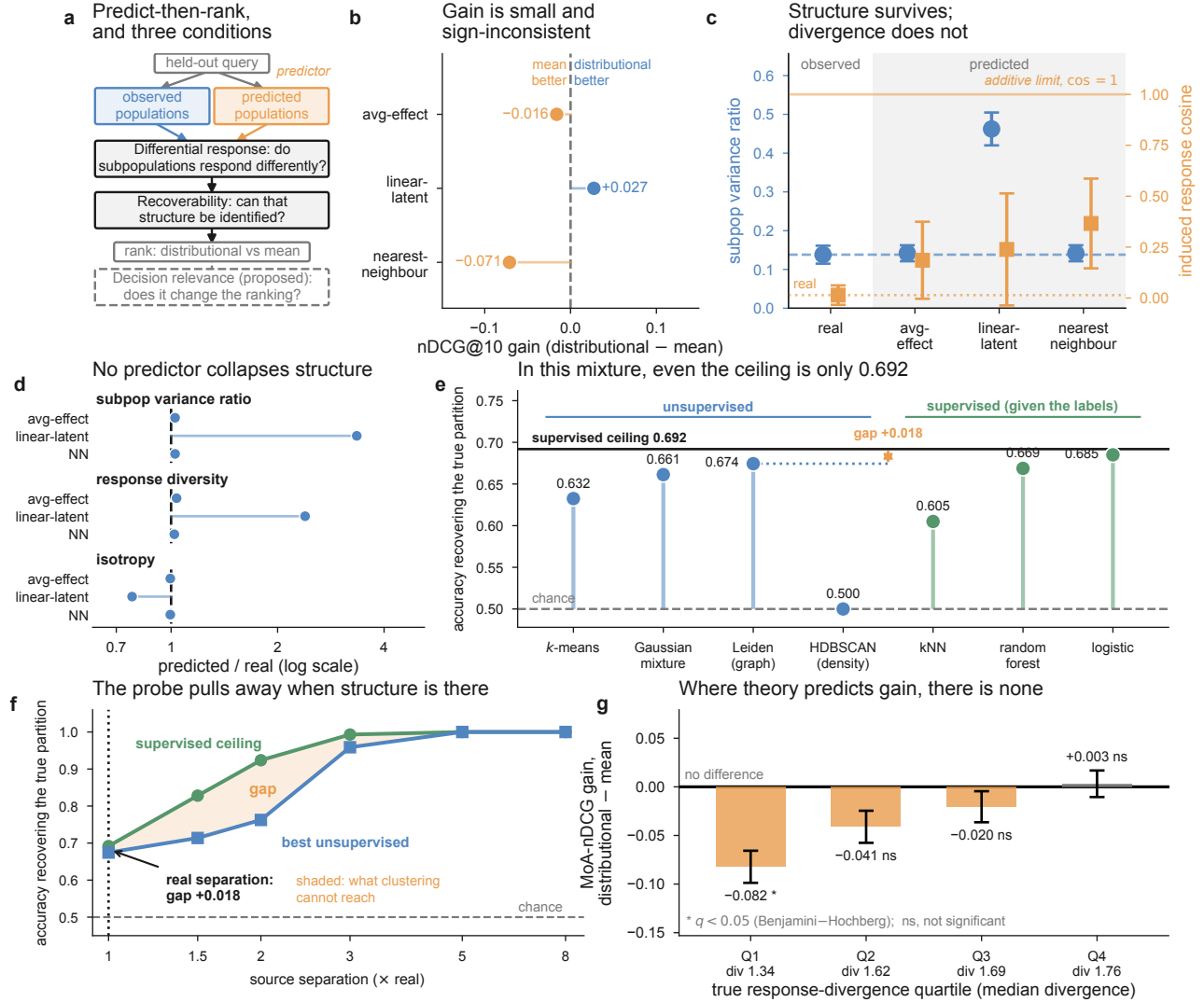


Figure 6: When can population-level information improve drug retrieval? **a**, Three requirements connect cellular heterogeneity to a change in candidate ranking. Differential response and recoverability are measured directly in this study; decision relevance is a downstream requirement proposed from the combined results and is shown separately. **b**, When candidate populations are generated by perturbation predictors rather than directly observed, the population-level advantage is small and inconsistent across the three predictor families tested (distributional minus mean nDCG@10: -0.071, -0.016 and +0.027). **c**, Differential response. Predicted populations can preserve heterogeneity in the starting cell population while failing to reproduce state-specific perturbation effects. Additive predictors necessarily assign the same perturbation response to every cellular state. **d**, Structure diagnostics of predicted candidate populations relative to the real population they replace, on a ratio axis with the reference at 1. The additive average-effect and nearest-neighbour predictors reproduce subpopulation variance, response diversity and isotropy almost exactly (ratios 0.99–1.04); the linear-latent model overshoots subpopulation variance and response diversity (3.35 and 2.39) and reduces isotropy (0.78). **e**, Recoverability in a controlled two-state mixture. The best unsupervised method reaches 0.674 compared with a supervised ceiling of 0.692; the low ceiling is partly explained by pooling multiple drugs into broad classes, whereas individual drug pairs reach a ceiling of 0.879. Natural tumour tissue shows a larger supervised–unsupervised gap (Fig. 5d). **f**, Increasing the separation of the constructed states causes the supervised ceiling to pull away from unsupervised recovery, confirming that the probe can detect additional recoverable information when it is present. **g**, Decision relevance is not implied by differential response alone. Mechanism-of-action gain does not increase into a significant positive advantage in the highest response-divergence quartile (mean +0.003, $q = 0.57$), whereas the lowest-divergence quartile is significantly negative.

retained as distributions. The central distinction is therefore not mean versus distributional retrieval, but information that is available versus information that changes the decision. Classical perturbation signatures and single-cell populations are thus better viewed as points on a continuum of retained response structure; the relevant question is how much of that structure must be preserved for the decision at hand.

This distinction has direct implications for artificial intelligence virtual cells^{1,2}. A virtual cell that accurately predicts an average perturbation response may still fail to provide the information needed for population-level intervention selection. In particular, a model that applies the same perturbation effect^{6,7} to every cell can preserve heterogeneity in the starting population yet, by construction, cannot generate state-specific differential responses. A downstream population-aware retriever cannot recover variation that the predictor never generated. Models intended to support intervention selection must therefore do more than predict mean responses: they must reproduce how perturbation effects vary across cellular states, and those differences must remain recoverable in the predicted population. We could not test whether satisfying this requirement is sufficient to improve cross-context retrieval because no predictor available here both transferred accurately across contexts and reproduced the observed degree of differential response. This gap defines a concrete target for virtual-cell modelling rather than a general verdict on existing model families.

Taken together, the results suggest three conditions under which population-level retrieval can add value beyond a mean signature. Cellular states must respond differently to a perturbation, those differences must be recoverable from the measured or predicted population, and the additional information must be consequential enough to change which candidate is preferred. The first two determine whether useful population-level information is available; the third determines whether it matters for the retrieval decision. This final requirement also clarifies what constitutes convincing validation. An external measurement is not automatically a neutral judge if its statistical form preferentially preserves the same structure as the retrieval method^{21,29}, so validation should ask whether the added cellular information changes an outcome relevant to the intended use and should compare against simple alternatives such as response magnitude²⁷ before attributing a gain to population structure itself. The functional readouts available here remain indirect and do not establish therapeutic utility. More broadly, richer cellular measurements justify richer retrieval methods only when the information they add is retained, recoverable and consequential for intervention choice.

Methods

Study design and data resources

This study asks when retaining the full population of single-cell perturbation responses improves retrieval beyond a conventional mean expression signature. The analyses were therefore organized around four distinct roles: observed-population retrieval, evaluation against biological and functional criteria, measurement of naturally occurring response heterogeneity, and large-scale assessment of the conditions under which population-level information can be useful. These roles were analysed separately rather than pooled into a single benchmark.

The primary observed-population retrieval analyses used SciPlex3¹⁵, comprising 276,325 cells across three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and

a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-context constructions because the same compound library is measured across all three cell lines. Frangieh melanoma Perturb-CITE-seq³² contains 218,023 RNA-profiled cells across Control, IFN γ and Coculture conditions with 239 CRISPR knockouts and 2,000 highly variable genes. We use this dataset as an independent perturbation-retrieval setting and, because matched surface-protein measurements are available, for the external evaluator-shape experiment described below. It is a genetic-perturbation test of the same retrieval formulation rather than a drug-retrieval benchmark. CD34+ haematopoietic progenitor data (GSE306429)³⁴ contain 33,984 cells and 36 compounds and are used as a negative-control setting for population-level retrieval.

Three additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021¹⁷ contains drug-treated patient-derived glioblastoma tissue and is used to test whether state-dependent responses and their recoverability are present in naturally heterogeneous tissue; no retrieval performance is reported from this cohort. Tahoe-100M¹⁶ is used as a larger-scale measurement of the same information requirements under intrinsic within-cell-line heterogeneity. We analyse plate 3 only, which comprises 4,158,278 cells across 50 cell lines and 93 compounds, using the released 2,304-gene log1p-transformed representation without refitting the preprocessing pipeline. Finally, GDSC2 dose-response profiles provide an external functional readout for drug-drug similarity and are joined to the SciPlex3 compound panel independently of the transcriptomic retrieval scores.

Observed responses, predicted responses, external functional measurements and natural-tissue analyses therefore answer different questions in this study. Results are not combined across these resources as though they represented independent replicates of one benchmark. Dataset scale, inclusion criteria, experimental role and query construction are summarized in Supplementary Table 1.

Preprocessing and construction of perturbation-response populations

Single-cell expression analyses were performed in log-transformed expression space using the processed gene sets defined for each dataset. SciPlex3, Frangieh and CD34+ were analysed in 2,000-gene representations. For SciPlex3 and Frangieh these genes were selected in this study rather than inherited from the released objects: counts were normalized to a total of 10^4 per cell and log1p-transformed, 2,000 highly variable genes were chosen by the Seurat criterion³⁵ on a subsample of cells, and the final matrices were then rebuilt from raw counts on those genes using all-gene size factors so that normalization is applied once. SciPlex3 was additionally downsampled to at most 120 cells per compound, cell line and dose, and 3,000 control cells per cell line. The CD34+ tensor was received in processed form and its feature-selection command is not recorded in this repository. For ZhaoSims2021, counts were normalized to a total of 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting. Dataset-specific filtering and minimum-cell requirements for individual analyses are given in the corresponding Methods sections below.

For a perturbation d measured in cellular context c , let

$$X_{d,c} = \{x_1^{(d,c)}, \dots, x_{n_{d,c}}^{(d,c)}\}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)}$$

denote the mean of the matched control population. The conventional perturbation signature is the control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}.$$

Mean-signature retrieval operates only on $\bar{d}_{d,c}$.

Population-level retrieval retains the corresponding control-referenced cellular responses,

$$P_{d,c} = \left\{ x_i^{(d,c)} - \mu_{0,c} \right\}_{i=1}^{n_{d,c}},$$

so that the mean of $P_{d,c}$ is exactly the conventional mean perturbation signature while variation among its cells remains available to the retrieval score. This construction makes the comparison between mean-signature and population-level retrieval differ in the response structure retained rather than in the underlying perturbation or control definition.

Controls were matched to the biological context in which the perturbation was measured. For analyses that explicitly compare responses between cellular states or compartments, each state is instead referenced to its own matched control population so that pre-existing differences between states cannot be mistaken for differential perturbation responses. This state-specific construction is used for the differential-response analyses in predicted populations, patient-derived glioblastoma and Tahoe-100M and is defined in the corresponding sections below.

Cell sampling was performed only after the response populations had been defined. Minimum cell numbers, per-condition caps and random seeds differ by experiment and are reported with the corresponding analysis rather than imposed globally. Protein measurements used for the evaluator-shape experiment were processed separately, including barcode-based RNA-protein matching and protein-specific normalization, as described below.

Population-to-population retrieval and signature baselines

For a query response population Q and a library of candidate response populations $\{P_d\}_{d=1}^D$, retrieval assigns each candidate a similarity or distance to the query and orders the candidate library accordingly. Population-level methods operate directly on the empirical response populations defined above, whereas signature-based methods first reduce each population to its control-referenced mean response. Query construction and candidate libraries are kept identical when comparing the two representations.

Population-level retrieval. Five population-level variants were evaluated: energy distance, MMD, sliced Wasserstein, coverage-mean and coverage-worst. Energy distance³⁶ was the primary score. The RBF-kernel maximum mean discrepancy (MMD)³⁷ uses a median-heuristic bandwidth evaluated at three scales (0.25, 1 and 4 times the median squared pairwise distance of the pooled sample), because bandwidths fixed in advance underflow in a 2,000-dimensional gene space. The sliced Wasserstein-1 dis-

421 tance³⁸ averages quantile-matched one-dimensional projections over 128 random directions, which ac-
 422 commodates unequal sample sizes. The final two variants are the two limits of the single subpopulation-
 423 coverage family defined below. All five rank candidates from smallest to largest distance to the query.
 424 Candidate populations were subsampled to a fixed maximum cell count with a fixed random seed be-
 425 fore any distance was computed (500 cells by default; 300 in the partial-observation experiments); a
 426 population left with fewer than two cells was assigned a fixed large penalty distance rather than silently
 427 dropped.

Subpopulation-coverage score. The query is partitioned into K components by k -means on the query cells ($k = 2$ throughout, three initializations, fixed seed), and each candidate cell is assigned to the component whose query centroid it most resembles in cosine similarity. Writing d_k for the base distance between the query cells and the candidate cells of component k , the coverage distance and its ranking score are

$$D_\beta = \frac{1}{\beta} \left(\log \sum_{k=1}^K e^{\beta d_k} - \log K \right), \quad S_\beta = -D_\beta.$$

428 The base distance d_k is one of energy, MMD or sliced Wasserstein, so coverage is a family built on the
 429 same kernels rather than an independent metric. A candidate that populates no cell of a query compo-
 430 nent receives the penalty distance for that component, which is how the score charges a candidate for
 431 leaving part of the query uncovered.

432 Three properties fix the interpretation of β . As $\beta \rightarrow 0$, $D_\beta = \text{mean}(d) + \frac{\beta}{2} \text{var}(d) + O(\beta^2)$ and the
 433 aggregate becomes the arithmetic mean over components; the implementation evaluates this expansion
 434 directly at small β , because the $(\log \text{sumexp} - \log K) / \beta$ form cancels catastrophically there. As $\beta \rightarrow \infty$,
 435 $D_\beta \rightarrow \max_k d_k$, the worst-matched component. For $K = 1$, $D_\beta = d_1$ for every β , so the coverage family
 436 contains the global distance exactly rather than as an approximation. Coverage-mean and coverage-
 437 worst are these two limits, and are the two variants used in the main analyses.

438 **Mean-signature baselines.** For each response population $P_{d,c}$, the corresponding mean signature is
 439 $\bar{d}_{d,c} = \text{mean}(P_{d,c})$. Mean-signature retrieval ranks candidates using either cosine similarity or Euclidean
 440 distance between these vectors. The implemented CMap-style cosine baseline applies cosine similarity to
 441 the same control-referenced mean differential-expression vectors and is therefore numerically identical
 442 to mean-cosine retrieval. The canonical L1000 weighted connectivity score (WTCS) is included as a
 443 separate signature-based baseline. WTCS uses rank-based enrichment of the mean-collapsed expression
 444 response and is related to, but not algebraically identical to, the cosine implementation used here.

445 To test whether the comparison depended on the original gene space, we also evaluated a PCA rep-
 446 resentation. PCA was fitted separately for each retrieval query, after which the same comparison was
 447 performed either between latent-space means or between the corresponding latent response popula-
 448 tions.

Connection between mean and population retrieval. To formalize the relationship between the two representations, we write a response population as

$$X = \mu + \lambda \epsilon, \quad E[\epsilon] = 0,$$

where μ is the mean response, ϵ represents within-population variation and λ controls its magnitude. As $\lambda \rightarrow 0$, the response population collapses to a point mass at μ , and population comparison therefore loses all information beyond the mean. Mean-signature retrieval is the mean-only endpoint of this variance-sensitivity construction.

This limiting argument does not imply that every population distance becomes algebraically identical to every conventional signature score. The exact identity used in this study is narrower: mean-cosine retrieval and the implemented CMap-style cosine baseline apply the same cosine operation to the same mean differential-expression vectors. WTCS remains a distinct rank-based signature score, and energy, MMD and sliced Wasserstein retain their own distance definitions. The continuum therefore refers to the amount of within-population response structure available to the scorer, not to an equivalence between otherwise different scoring functions.

Response-matching and partial-observation evaluation

Observed-population retrieval. When complete candidate response populations were available, retrieval was evaluated as source identification. For each query, the candidate library contained the perturbation that generated the query response together with competing perturbations defined by the corresponding task. Hit@1 is the fraction of queries for which the true source perturbation is ranked first. Comparisons between population-level and mean-signature retrieval use the same queries and candidate libraries. Aggregate Hit@1 values are reported as unweighted macro-means over the seven task-setting combinations used in the observed-population analysis; query-weighted averages are reported as a sensitivity analysis so that the largest task does not determine the primary summary.

Partial-observation protocol. We additionally evaluated retrieval when only part of each candidate response population was available to the retriever. This setting was designed to separate the information used to construct a ranking from the criterion subsequently used to evaluate that ranking. For each query, the population-level and mean-signature methods rank the same candidate library once; the resulting rankings are then retained unchanged while different evaluation criteria are applied.

Three splits construct the partial observation, all on SciPlex3 and all leakage-safe³⁹ in the sense that the hidden material never enters the candidate library. In *leave-drug-out*, one drug is hidden, the query is that drug’s own treated population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by k -means with $k = 2$ on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate’s welfare is the aggregate over query states s of its negative energy distance to that state, $U(d) = \text{agg}_s (- \mathcal{E}(P_d, Q_s))$ with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate $\hat{d}$ is

$$\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0,$$

where $\mathcal{L}$ is the observed library, so the reference is the best candidate actually available rather than an unobservable optimum. Minority-state coverage is reported on the same rankings as the cosine between the candidate’s mean response and the mean response of the query’s minority state, rescaled from $[-1, 1]$ to $[0, 1]$. Regret is zero when the top-ranked candidate is optimal under that objective and increases as the selected candidate becomes less favourable. We report regret reduction relative to mean-signature retrieval, such that a positive value indicates lower regret under this response-matching criterion. This evaluation is intentionally closely aligned with the population-level information used by the retriever and is therefore interpreted as evidence of response fidelity rather than as an independent functional outcome.

Predict-then-rank evaluation. The predictor-generated retrieval reported in Fig. 6b factorizes the pipeline the field actually uses into a forward predictor and a retrieval rule, so that the retrieval layer can be varied with the predictor held fixed. Queries are cross-line: for each of the three SciPlex3 cell-line pairs (K562–A549, A549–MCF7, K562–MCF7) and each of ten drugs selected by a cross-context divergence probe, the query is that drug’s response blended across the two lines at mixing fraction $\alpha \in \{0.5, 0.7, 0.9\}$ over 200 cells, with eight seeds, giving 720 queries. The candidate library is the query’s own drug together with up to 30 further drugs shared by both lines; the query’s drug is the single relevant candidate. Each predictor is asked to imagine every candidate’s response in the query’s real blended control cells, mixed at the same α , so no predictor sees the within-query line structure. The imagined populations are then ranked by four retrieval rules: mean cosine, R^2 of the predicted against the observed mean delta, energy distance and coverage. Performance is nDCG@10 with binary relevance, and the quantity plotted is the distributional minus the mean-cosine value for the same predictor, so each predictor is its own control.

Task-proximal biological evaluation. The same fixed rankings were next evaluated using mechanism-of-action (MoA) recovery and minority-state coverage. MoA recovery was quantified by normalized discounted cumulative gain (nDCG)⁴⁰, which rewards candidates sharing the query perturbation’s annotated mechanism when they occur near the top of the ranked list. The comparison is defined only when the candidate library contains members of the query mechanism class. Of the 765 partial-observation queries, 600 satisfy this requirement. The remaining 165 queries—120 leave-MoA-out queries and 45 partial-library queries—are excluded from MoA analyses rather than assigned an artificial zero, because the relevant mechanism is absent from the candidate library and recovery is therefore undefined.

Minority-state coverage asks whether the top-ranked candidate better preserves the response of the less represented cellular state in the controlled heterogeneous-population construction. As with MoA recovery, this criterion is applied to rankings that have already been generated; the evaluator does not alter the query, the retrieval score or the candidate ordering. Differences between population-level and mean-signature retrieval are therefore attributable to how the same rankings score under the biological criterion rather than to refitting or re-ranking.

Real-data projection. To examine whether the minority-state result extends beyond the controlled partial-observation setting, we evaluated the top-ranked population-level and mean-signature candi-

521 dates across 239 real-data tasks. These comprised 90 constructed cross-cell-line SciPlex3 tasks, 90 within-
 522 cell-line SciPlex3 tasks, 23 Frangieh tasks, 12 CD34+ tasks and 24 tasks using predicted candidate pop-
 523 ulations. For each task we report the difference in minority-state coverage between the population-level
 524 and mean-signature selections.

525 The 239 tasks are treated as a distribution of task-level differences rather than as independent bio-
 526 logical replicates of a single experiment. In particular, the constructed SciPlex3 tasks share data source,
 527 preprocessing and candidate structure, and the three SciPlex3 cell lines constitute leave-one-cell-line-out
 528 variation within one experiment rather than leave-one-dataset-out generalization. Thresholded counts
 529 of “distributionally dominant” tasks are therefore not used as a primary result; the continuous difference
 530 in coverage is reported instead.

531 **Controlled synthetic benchmark**

HIR-Bench was used as a controlled model of when within-population structure can change the
 preferred retrieval candidate. The benchmark generates heterogeneous retrieval tasks from a two-
 subpopulation latent welfare model with tunable subpopulation balance and conflict between
 subpopulation-specific preferences. Under this construction, the preferred candidate changes at the
 analytical boundary

$$\alpha^* = \frac{B}{A + B},$$

532 where the notation follows the latent welfare specification of the generator. The boundary therefore
 533 provides a known-ground-truth distinction between regimes in which the population mean is sufficient
 534 to preserve candidate preference and regimes in which resolving the two subpopulations can change
 535 that preference.

536 Each parameter configuration was instantiated under three information conditions: fully observed
 537 response populations, predicted populations containing only mean-level response information, and pre-
 538 dicted populations retaining population structure. The full benchmark comprises 672 parameter config-
 539 urations evaluated over 20 random seeds, giving 13,440 instances. HIR-Bench is used to establish the
 540 existence and observability of a controlled preference-changing regime; it is not treated as a generative
 541 model of any specific biological dataset.

542 To ask whether entry into this regime could be recognized from information available at retrieval
 543 time, we fitted logistic-regression classifiers to predict whether population structure changes the pre-
 544 ferred candidate. The observable feature set contains only statistics that can be computed from the
 545 query and candidate response populations: subpopulation-variance ratio, isotropy index and response
 546 diversity. For comparison, an oracle-derived feature set contains top- k disagreement and Kendall conflict
 547 scores computed from the latent utility matrix used by the generator. These oracle-derived quantities are
 548 not available to a retrieval system and are included only as a controlled comparison.

549 Cross-validation was grouped by the parameter pair $(\alpha, \text{conflict})$, because the retrieval-flip label is
 550 determined by these two variables. This gives 28 label-determining parameter cells, which are treated
 551 as the effective independent units. Instances generated from the same parameter cell are therefore never
 552 divided between training and test sets. Because each held-out parameter cell contains a single outcome
 553 class by construction, AUC is computed from the pooled out-of-fold predictions rather than within indi-
 554 vidual folds.

We additionally examined whether the synthetic regime boundary could be transferred to the real-data tasks. The method-performance layer required for this projection was available only on a reduced grid; the resulting classifier was compared against a constant majority-class baseline on the real-data projection (Fig. 5b). On the basis of that comparison we do not use HIR-Bench to assign real biological datasets to a population-favourable or mean-sufficient regime. In the main text, it serves only as a controlled demonstration that population structure can, in principle, become consequential for candidate choice.

External functional evaluation using GDSC2

To test whether transcriptomic retrieval rankings capture information that extends beyond the single-cell expression measurements used to construct them, we evaluated the SciPlex3 rankings against drug-response profiles from the Genomics of Drug Sensitivity in Cancer (GDSC2) resource^{30,31}. The external criterion was defined as similarity between the functional response profiles of two drugs rather than as absolute drug potency, because the retrieval task asks which candidate produces a response most similar to the query rather than which candidate is most cytotoxic.

SciPlex3 compounds were matched to GDSC2 by normalized drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. GDSC1 and GDSC2 measurements were not pooled because they derive from different assay and dose-range settings. For each matched compound, its functional profile was the vector of fitted dose-response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments—A549, K562 and MCF7—were excluded from these profiles, so the functional evaluation was performed in biological contexts not used to construct the RNA rankings.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line effect before calculating drug-drug similarity. For cell line l and compound d , the centred response was

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in cell line l , rather than only the 35 compounds overlapping SciPlex3. Functional similarity between query drug q and candidate drug c was then defined as

$$F(q, c) = \rho_s \left(\widetilde{\text{AUC}}_{q,\bullet}, \widetilde{\text{AUC}}_{c,\bullet} \right),$$

where ρ_s denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10 μM response population of the held-out compound served as the query, and the remaining matched compounds formed the candidate library, giving 34–35 candidates per query. Each RNA-based retrieval rule produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the external functional ordering $F(q, c)$ was quantified by Spearman correlation across candidates.

The primary RNA-derived rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q, c) = - \left| \|\bar{d}_q\| - \|\bar{d}_c\| \right|,$$

which prefers candidates whose overall transcriptional response magnitude is similar to that of the query but uses no within-population structure. A query-independent magnitude score $\|\bar{d}_c\|$ was retained as a diagnostic control. An uncorrected mean-cosine score based on treated means without matched-control subtraction and a potency-matching diagnostic were analysed as additional sensitivity checks rather than as primary retrieval baselines.

To estimate the component of the energy-based association not explained by response magnitude, we computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores and functional similarities were rank-transformed, each was residualized with respect to $M(q, c)$, and the correlation between the two residuals was reported as the magnitude-adjusted association.

Per-query Spearman correlations are summarized by their median within each SciPlex3 cell line. Queries from the same cell line share the same candidate library and are therefore not treated as independent biological replicates. The three cell lines provide the higher-level analysis unit; query-level paired tests are reported only as descriptive analyses where indicated.

Protein-based evaluator-shape experiment

Frangieh Perturb-CITE-seq³² provides matched RNA and surface-protein measurements from the same cells, allowing the statistical form of an external evaluator to be changed while holding its biological measurement fixed. RNA and protein matrices were joined by cell barcode before analysis; row position was not used because the released modality-specific matrices differ in cell number and ordering. Of the 24 measured antibodies, four isotype controls were removed, leaving 20 surface-protein features. Protein counts were centred-log-ratio normalized across antibodies within each cell.

All analyses were performed separately within the Control, IFN γ and Co-culture conditions to avoid mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-matched cells were retained, and populations were subsampled to at most 250 cells using a fixed random seed.

Protein-based evaluators. For knockout k in immune condition c , let $\bar{d}_{k,c}^{\text{prot}}$ denote its mean control-referenced protein response and let $D_{k,c}^{\text{prot}}$ denote the corresponding set of per-cell protein responses. The mean-based evaluator was defined as

$$E_{\text{mean}}(q, d) = \cos \left(\bar{d}_{q,c}^{\text{prot}}, \bar{d}_{d,c}^{\text{prot}} \right).$$

The population-based evaluator was defined from the same cells and protein measurements as

$$E_{\text{pop}}(q, d) = -\mathcal{E} \left(D_{q,c}^{\text{prot}}, D_{d,c}^{\text{prot}} \right),$$

where $\mathcal{E}$ is the energy distance. The negative sign converts distance to similarity so that larger values

614 indicate a closer match under both evaluators.

615 For both definitions, protein responses were referenced to the matched control population from the
616 same immune condition. Thus,

$$\bar{d}_{k,c}^{\text{prot}} = \text{mean} \left(X_{k,c}^{\text{prot}} \right) - \mu_{0,c}^{\text{prot}},$$

617 and

$$D_{k,c}^{\text{prot}} = \left\{ x_{i,k,c}^{\text{prot}} - \mu_{0,c}^{\text{prot}} \right\}_{i=1}^{n_{k,c}}.$$

618 Using the same matched control for the two evaluator forms ensures that their difference arises from
619 whether protein responses are summarized by their mean or retained as a cellular population, rather
620 than from a change in biological reference.

621 **Fixed RNA-derived rankings.** The RNA-derived rankings were computed once and then evaluated
622 against both protein criteria. The primary rankings were energy-distance population retrieval and
623 control-referenced mean-cosine retrieval. The query populations, candidate libraries and RNA-derived
624 candidate scores were identical under the two protein evaluators, and no protein measurement was
625 available to any retrieval method. The experiment therefore changes only the statistical representation
626 used to judge the same RNA-derived ranking.

627 **Response-magnitude control.** Because a population distance can also encode differences in overall re-
628 sponse magnitude, we included a query-dependent scalar control that uses no cell-to-cell distributional
629 information. For query q and candidate d , magnitude matching was defined as

$$M(q, d) = - \left| \left| \bar{d}_{q,c}^{\text{RNA}} \right| \right| - \left| \left| \bar{d}_{d,c}^{\text{RNA}} \right| \right|.$$

630 This ranking favours candidates with a transcriptional response magnitude similar to that of the
631 query while discarding all within-population structure. The same magnitude ranking was evaluated
632 against both protein criteria.

633 For each query, agreement between an RNA-derived ranking and a protein-based evaluator was
634 quantified by Spearman correlation across candidate knockouts. Query-level correlations were sum-
635 marized by their median within each immune condition. The three immune conditions were reported
636 separately and treated as the higher-level analysis unit; queries within a condition share candidate struc-
637 ture and were not interpreted as independent biological replicates.

638 Differential-response analysis of predicted populations

639 We next asked whether perturbation predictors preserve the state-specific response variation that
640 population-level retrieval is intended to exploit. These analyses were not designed as a benchmark of
641 perturbation-prediction models. Instead, the predictors were used as alternative sources of candidate
642 response populations to test which forms of response structure remain available after prediction.

643 **Predict-then-rank response sources.** Three predictor families were used in the predict-then-rank anal-
644 ysis. The average-effect predictor estimates a mean perturbation effect for each drug and applies that

effect to cells in the query context. The nearest-neighbour predictor transfers the perturbation effect from the training context whose untreated control state is most similar to the query context. The linear-latent predictor implements latent arithmetic in which a perturbation shift is applied in a learned latent representation before decoding cells back to expression space. These models were used only to generate candidate populations for retrieval and were not interpreted as representative benchmarks of the broader virtual-cell model class.

To construct a predicted population, the predicted perturbation effect was applied to the real control cells of the query context one cell at a time. Thus, a predicted population retained the cell-to-cell variation already present in the untreated population whenever the predictor permitted that variation to propagate through the model. We did not use synthetic Gaussian clouds of the form control mean $+\delta + \epsilon$ for differential-response analyses, because such a construction imposes its own population geometry and can obscure whether the predictor itself generates state-dependent effects.

The reported predict-then-rank experiment was fitted using the full SciPlex3 matrix, and the cell lines contributing to the query populations were not excluded from predictor fitting. These results should therefore be interpreted as an in-sample ceiling on the candidate-generation step rather than as evidence of cross-context predictive generalization.

Within-context definition of differential response. Differential response was measured within a single cellular context so that variation between contexts could not be mistaken for state-specific perturbation effects. For each SciPlex3 context, up to 400 untreated control cells were represented by 30-component PCA and partitioned into two cellular states using k -means with $k = 2$. Subpopulation sizes and silhouette scores were retained as diagnostics of the underlying state partition.

For perturbation d , each predictor was applied separately to the control cells belonging to the two states. The induced response of state k was defined as

$$r_{d,k} = \text{mean}(\hat{X}_{d,k}) - \text{mean}(X_{0,k}),$$

where $X_{0,k}$ denotes the untreated cells assigned to state k and $\hat{X}_{d,k}$ denotes their predicted post-perturbation profiles. Differential response was summarized by the cosine between the two state-specific response vectors,

$$R_d = \cos(r_{d,0}, r_{d,1}).$$

A value near 1 indicates that the two cellular states are predicted to respond in nearly the same direction, whereas lower values indicate increasingly state-specific perturbation responses. Real treated cells from the same context, partitioned according to the same control-derived state definition, were analysed identically and provide the observed reference.

Additive-response limit. The limiting behaviour of an additive predictor follows directly from its model form. If the same perturbation effect δ_d is applied to every cell,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

then the induced response is identical in every cellular state,

$$r_{d,0} = r_{d,1} = \delta_d,$$

and therefore

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

An additive predictor may therefore preserve substantial heterogeneity in the untreated population while generating no state-specific variation in the perturbation effect itself. This distinction is why population heterogeneity and differential perturbation response are measured separately.

Latent-model reconstruction reference. For predictors that reconstruct the input before applying a latent perturbation shift, the measured response depends on the reference used to define the perturbation effect. Let $P(C_k)$ denote the model reconstruction of control state C_k , and let $W\delta_z$ denote the decoded latent perturbation shift. Referencing the predicted population to the observed control cells gives

$$r_k^{\text{obs}} = \text{mean}(P(C_k)) - \text{mean}(C_k) + W\delta_z,$$

whereas referencing it to the model's own reconstructed control gives

$$r_k^{\text{model}} = W\delta_z.$$

The first quantity includes state-dependent reconstruction error; the second isolates the perturbation effect represented by the model. Both references were therefore retained for latent predictors. The additive-limit identity above applies to the model-referenced response, for which a shared latent shift necessarily yields a cosine of 1.

Additional predictor diagnostics. The within-context differential-response analysis also included a deployed scGen model and a non-parametric optimal-transport mapping to test whether predictors not restricted to the simple gene-space average-effect construction generated response variation closer to that observed in real treated populations. These models were evaluated with the same state partition and induced-response definition described above.

The scGen model is the published implementation⁶ (`scgen.SCGEN` on `scvi-tools`⁴¹), trained on the SciPlex3 tensor with the perturbation label as the batch key and the cell line as the label key, 30 latent dimensions, batch size 512, at most 40 epochs with early stopping after 15 epochs without improvement, and a fixed seed. Its own latent-arithmetic rule is used verbatim: one latent delta per perturbation, estimated as the mean latent of treated cells minus the mean latent of control cells, and a prediction is formed by encoding the query context's control mean, adding that delta and decoding. Because the decoder is nonlinear, a single latent delta decodes to a different gene-space response for each cell, so the induced-response cosine is not pinned to 1 by construction as it is for the linear-latent control.

The optimal-transport map is an entropic (Sinkhorn) coupling³³ solved with POT⁴² between control and treated cells, with uniform marginals, regularization 0.05, at most 500 iterations, and a cost matrix

rescaled by its maximum. It is solved in a 30-component PCA latent fitted unsupervised on at most 20,000 cells, because in the full 2,000-gene space all pairwise costs are nearly equal after rescaling and the plan carries no signal; the gene-space variant did not clear the learning check and was not used. Each fitting context is centred on its own control mean before the coupling is solved, so the map transports response rather than baseline identity. The map is fitted only on contexts other than the query's, so the query context's treated cells are never seen. A barycentric projection of the coupling gives a per-source-cell displacement, which is extended to the query's real control cells by nearest-source matching in the same centred latent space and decoded through the PCA inverse transform.

Both models are held to a pre-specified learning check: a predictor whose transferred mean-delta cosine falls below 0.30 is declared void and is not interpreted. scGen (0.524) and the optimal-transport map (0.352) pass this criterion. Each predictor's induced response is measured against its own zero-effect output rather than against real control cells, which cancels the reconstruction bias that latent models otherwise contribute to this quantity. Values are means over 30 cell-line and drug combinations (three cell lines, ten drugs, 400 control cells per context split into two states).

No model evaluated here was used to establish a general ranking of perturbation-prediction architectures. The analysis tests a narrower requirement: whether a predicted population contains state-specific perturbation effects that could, in principle, be exploited by a downstream population-aware retriever.

Recoverability of state-specific responses

Differential responses can contribute to population-level retrieval only if the corresponding cellular structure can be recovered from the observed response population. We therefore measured recoverability by comparing unsupervised partitioning with a supervised estimate of how well the same states could be distinguished when their identities were known. This comparison separates failure of an unsupervised method from absence of discriminative information in the measured representation.

Controlled two-state response mixtures. The primary recoverability experiment used K562 cells treated at 10 μ M with compounds from HDAC and JAK perturbation classes. Response populations from the two sources were combined to form mixtures with known two-state composition. Between 50 and 400 cells were sampled from each source over 10 random seeds. Ground-truth source identity was used only for evaluation and for the supervised ceiling; it was not available to the unsupervised procedures.

All methods were evaluated using best-permutation accuracy so that supervised and unsupervised procedures shared the same outcome scale. For a clustering assignment z_i and source label y_i , cluster identities were matched to source identities after fitting, and accuracy was computed as

$$A = \max_{\pi} \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i = \pi(z_i)],$$

where π denotes the permutation of cluster labels that maximizes agreement with the known sources. For methods returning more than two clusters, each cluster was assigned to its majority source label. HDBSCAN noise points were treated as an additional cluster before label matching. This convention favours rather than penalizes over-partitioning and therefore provides a conservative comparison with the supervised ceiling.

743 **Unsupervised recovery.** Four unsupervised procedures were evaluated: k -means, Gaussian-mixture
744 modelling, Leiden clustering⁴³ and HDBSCAN⁴⁴. The first two methods were given the known number
745 of source populations, $k = 2$, whereas Leiden and HDBSCAN determined their own partition structure.
746 Unless otherwise stated, clustering was performed in a 30-component PCA representation fitted to the
747 complete response mixture. The best unsupervised accuracy across the evaluated methods was retained
748 as the recoverable structure extracted without source labels.

749 **Supervised ceiling.** To estimate how much source information was present in the same expression rep-
750 resentation, we trained logistic regression, random forest and k -nearest-neighbour classifiers using the
751 known source labels. Models were evaluated by five-fold stratified cross-validation, and the supervised
752 ceiling for each mixture was defined as the best out-of-fold accuracy among the three classifiers.

753 Because the clustering procedures operate on a PCA representation fitted to the complete mixture, we
754 report two supervised references. In the leak-free reference, PCA was re-fitted within each training fold
755 and test cells were projected without contributing to the fitted representation. In the matched reference,
756 the classifiers received the same transductive PCA representation as the unsupervised methods; PCA
757 itself does not use source labels. The matched representation provides the primary like-for-like compar-
758 ison between supervised and unsupervised recovery, whereas the in-fold representation is retained as a
759 sensitivity analysis.

760 Recoverability was summarized by the paired difference

$$G = A_{\text{sup}} - A_{\text{unsup}},$$

761 where A_{sup} is the supervised ceiling and A_{unsup} is the best unsupervised accuracy for the same mix-
762 ture. A small value of G indicates that little discriminative information remains beyond what unsu-
763 pervised partitioning already extracts. A larger value indicates that additional response information is
764 present in the representation but is not recovered by the unsupervised procedure. The gap is therefore
765 interpreted jointly with the absolute supervised ceiling rather than as a stand-alone measure.

766 **Drug-pair resolution.** The pooled HDAC-versus-JAK construction combines cells from several com-
767 pounds within each class and therefore introduces within-class variation unrelated to the distinction
768 between two individual drug responses. We consequently repeated the analysis at drug-pair resolution.
769 Individual HDAC-versus-JAK drug pairs were analysed separately in K562 at $10 \mu\text{M}$ using the same
770 representation, unsupervised methods, supervised probes and best-permutation accuracy. Twenty-four
771 drug pairs were sampled with a fixed seed from the 513 available pairs. This analysis tests whether
772 an apparently low recoverability ceiling reflects genuinely weak response separation or the additional
773 heterogeneity introduced by pooling several drugs into broad mechanism classes.

774 **Sensitivity to state separation.** To verify that the recoverability analysis can detect additional infor-
775 mation when it is present, we constructed a positive-control series in which separation between the two
776 source populations was increased by a factor

$$s \in \{1, 1.5, 2, 3, 5, 8\}.$$

777 The transformation was defined once from the full source populations and then applied before cell
778 subsampling, so the separation manipulation was not fitted to the particular cells used for evaluation.
779 The unmodified data correspond to $s = 1$. Increasing s provides a controlled test of whether the su-
780 pervised ceiling separates from unsupervised recovery as the two response states become increasingly
781 distinguishable.

782 The same supervised-versus-unsupervised framework was subsequently applied to naturally het-
783 erogeneous glioblastoma responses and to Tahoe-100M, with dataset-specific sampling and partition
784 definitions described below. The recoverability analysis therefore measures an information requirement
785 rather than a particular clustering algorithm: whether state-specific response information is present in
786 the population, and how much of that information remains unextracted without labels.

787 **Natural heterogeneity and large-scale validation**

788 We next examined the information requirements for population-level retrieval in datasets in which cellu-
789 lar heterogeneity was not created by mixing experimentally distinct source populations. Patient-derived
790 glioblastoma was used to test the requirements in naturally heterogeneous tissue, and Tahoe-100M was
791 used to assess the same quantities at substantially larger scale. Neither dataset was used as an additional
792 retrieval benchmark.

793 **Patient-derived glioblastoma.** The ZhaoSims2021 dataset contains acute-slice cultures and surgical
794 biopsies from ten glioblastoma patients exposed to six compounds: Ana-12, Ispenisib, RO4929097,
795 Tazemetostat, etoposide and panobinostat. Expression counts were normalized to 10^4 per cell, log1p-
796 transformed and reduced to 2,000 highly variable genes. Because only one patient received more than
797 two compounds, the cohort does not support a sufficiently replicated leave-one-drug-out retrieval
798 benchmark. We therefore use it only to measure state-specific response, recoverability and the extent to
799 which population-level similarity differs from compartment-specific response ordering.

800 **Cellular-compartment assignment.** The released dataset does not provide resolved cell-type annota-
801 tions for the analysed cells. Compartments were therefore assigned from canonical marker sets. Marker
802 scores were computed from the full log-normalized expression matrix before highly variable gene selec-
803 tion. A cell was assigned to a compartment only when that compartment had the highest marker score,
804 the mean expression of its defining markers was at least 0.25, and its marker-score z-score exceeded the
805 second-highest compartment by at least 0.25. Cells failing either criterion were labelled unassigned and
806 excluded.

807 This procedure retained three compartments that passed marker-expression validation: malignant
808 glioma cells, tumour-associated myeloid cells and oligodendrocytes. Overall, 39.9% of cells were ex-
809 cluded as unassigned. Sensitivity analyses varied both assignment thresholds between 0.10 and 0.50
810 and gave similar differential-response and recoverability estimates; these analyses are reported in the
811 Supplementary Information. Compartment labels were used only to define and evaluate the biological
812 states and were never supplied to an unsupervised recovery procedure.

813 **Differential response in natural tissue.** State-specific perturbation responses were measured between
814 malignant and myeloid compartments. For patient p , drug d and compartment k , the response vector

815 was

$$r_{p,d,k} = \text{mean} \left(X_{p,d,k}^{\text{treated}} \right) - \text{mean} \left(X_{p,k}^{\text{control}} \right).$$

816 Only patient–drug combinations with at least 50 treated cells in both compartments were included.
817 Controls were matched within both patient and compartment. Differential response was summarized as

$$R_{p,d} = \cos \left(r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}} \right).$$

818 This construction removes pre-existing differences between patients and cellular compartments be-
819 fore comparing the directions of their drug-induced responses.

820 **Recoverability in natural tissue.** Recoverability was evaluated as a drug-response discrimination
821 problem rather than as a cell-type classification problem. For each eligible patient and compartment,
822 cells treated with drug a were distinguished from cells treated with drug b . Each drug contributed
823 at least 50 cells and was capped at 150 cells per split; five random seeds were used. This yielded 36
824 within-patient drug-pair splits across four patients and two cellular compartments.

825 The same protocol described above for controlled mixtures was then applied: unsupervised meth-
826 ods operated in a common 30-component PCA representation, source recovery was evaluated by best-
827 permutation accuracy, and the supervised ceiling was defined as the best five-fold cross-validated ac-
828 curacy among logistic regression, random forest and k -nearest neighbours. The primary recoverability
829 gap was computed using the matched PCA representation shared by the supervised and unsupervised
830 procedures,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

831 for drug-pair split j . The reported cohort-level gap is the median of the paired values $\{G_j\}$ rather
832 than the difference between the marginal supervised and unsupervised medians.

833 Because the 36 splits are not statistically independent, uncertainty was estimated at the patient level
834 rather than by treating drug pairs as independent replicates. Cluster bootstrap resampling was per-
835 formed over patients, and the analysis was additionally repeated after excluding the patient contribut-
836 ing the largest number of drug-pair comparisons. Fixed supervised–unsupervised method pairs were
837 also examined as a sensitivity analysis to the use of the best-performing method within each family.

838 **Proxy for downstream decision relevance.** The glioblastoma cohort does not contain enough drug
839 coverage to test whether population-level retrieval directly changes the preferred candidate. We there-
840 fore quantified a more limited property: whether drug similarity at the whole-population level preserves
841 the ordering observed within a specific cellular compartment.

842 For each pair of drugs measured in the same patient, we first computed the cosine similarity be-
843 tween their population-mean response signatures and the cosine similarity between their malignant-
844 compartment response vectors. Agreement between these two similarity structures was summarized by
845 Spearman correlation across drug pairs.

846 Because the whole-population mean contains malignant cells and therefore shares observations with
 847 the malignant-compartment response, we additionally used a disjoint comparison. For drug pair (a, b) ,
 848 myeloid-response similarity was defined as

$$S_{a,b}^{\text{myeloid}} = \cos(r_{a,\text{myeloid}}, r_{b,\text{myeloid}}),$$

849 and malignant-response similarity as

$$S_{a,b}^{\text{malignant}} = \cos(r_{a,\text{malignant}}, r_{b,\text{malignant}}).$$

850 We then computed

$$\rho_{\text{disjoint}} = \rho_S(S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}}),$$

851 using only drug pairs available within the same patient. The two sides of this comparison contain no
 852 shared cells and each compartment is referenced to its own matched control. This statistic is interpreted
 853 as a proxy for the amount of candidate-ordering headroom available to population structure, not as a
 854 direct measure of retrieval performance. No pass-fail threshold was applied.

855 **Large-scale validation in Tahoe-100M.** Tahoe-100M¹⁶ was used to re-measure the same information
 856 requirements under unconstructed within-cell-line heterogeneity at larger scale. We analysed plate 3,
 857 which contains a complete grid of 50 cell lines and 93 compounds at one concentration per compound,
 858 together with a DMSO_TF vehicle condition in every cell line. The released plate contains 4,158,278 cells
 859 with a median of 648 cells per cell-line-compound condition. We used the released 2,304-gene log1p-
 860 transformed representation without additional feature selection or representation fitting.

861 Two independently defined cellular-state partitions were evaluated. The first used the released cell-
 862 cycle annotation, comparing G1 with G2M cells and excluding S-phase cells as an intermediate state.
 863 The second was defined by fitting k -means with $k = 2$ to the control cells of each cell line and assigning
 864 treated cells to the nearest control-derived centroid. The latter partition is therefore fixed from the un-
 865 treated state before any drug-response measurement is examined. Both partitions were analysed without
 866 selecting between them according to downstream performance.

867 For each eligible drug-cell-line condition, the response of state k was defined relative to its own
 868 state-matched controls,

$$r_{d,c,k} = \text{mean}(X_{d,c,k}^{\text{treated}}) - \text{mean}(X_{c,k}^{\text{control}}).$$

869 Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos(r_{d,c,0}, r_{d,c,1}).$$

870 A condition was included only when at least 50 cells were available in each required treated and
 871 control state.

872 Tahoe recoverability was evaluated using drug identity rather than the state labels used above.
 873 Within each cell line, 20 random pairs of compounds were sampled, with balanced samples of up to

300 cells per compound. Pairs were required to contain at least 100 cells for each compound. The supervised reference was five-fold cross-validated logistic regression. The unsupervised arm included k -means in gene space, k -means and Gaussian mixtures in 50-dimensional PCA space, and Leiden clustering with its resolution reduced until two communities were returned. All methods were evaluated by best-permutation accuracy, and the recoverability gap was defined as above.

Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses separately in the two non-overlapping cellular states. Spearman correlation between the two resulting drug–drug similarity matrices was then calculated,

$$\rho_c = \rho_s \left(S_{a,b}^{(0)}, S_{a,b}^{(1)} \right).$$

High correlation indicates that the two cellular states preserve similar relative ordering among drug responses, whereas lower correlation leaves greater scope for state-specific information to change candidate preference. As in the glioblastoma analysis, this quantity does not directly test whether population-level retrieval improves a drug ranking.

Tahoe-100M was therefore used only to measure differential response, recoverability and this decision-relevance proxy at larger scale. No retrieval algorithm was run on Tahoe-100M, and no Hit@ k , regret or external functional retrieval metric was derived from this dataset.

Statistical analysis

Statistical units were defined by the experimental design rather than by the number of cells or query–candidate comparisons. Cells contribute to the estimation of a perturbation-response population but are not treated as independent biological replicates of retrieval performance⁴⁵. Likewise, repeated queries sharing the same candidate library or biological context are interpreted at the level of that shared experimental unit where appropriate.

Paired comparisons between retrieval methods evaluated on the same queries were performed using two-sided Wilcoxon signed-rank tests. Comparisons between independent groups used Mann–Whitney U tests, and monotonic associations were quantified using Spearman rank correlation. Where the same hypothesis was examined across multiple pre-specified strata, P values were adjusted using the Benjamini–Hochberg false-discovery-rate procedure⁴⁶. Effect sizes are reported together with significance tests, and statistical significance is not interpreted as evidence of practical importance when the corresponding effect is small relative to the operating range of the metric.

For the partial-observation analyses, comparisons between population-level and mean-signature retrieval were paired within query because the two methods were evaluated on the same query and candidate library. Mechanism-of-action analyses were restricted to the 600 queries for which the relevant mechanism class remained represented in the candidate library; undefined queries were excluded rather than encoded as zero. Stratified analyses of response divergence used Benjamini–Hochberg correction across strata. Where observed effects were small, achieved power and the sample size required to reach 80% power were reported to distinguish weak evidence from a precisely estimated negligible effect.

For the GDSC2 functional analysis, Spearman correlations were first computed separately for each query across its candidate drugs and then summarized within each SciPlex3 cell line. Queries from the

912 same cell line share a candidate library and are therefore not independent biological replicates. Query-
913 level paired tests are reported only where indicated and are explicitly treated as nominal; the three Sci-
914 Plex3 cell lines provide the higher-level analysis unit for the functional comparison. The protein-based
915 evaluator-shape experiment was analysed analogously: per-query rank correlations were summarized
916 within each of the Control, IFN γ and Co-culture conditions, and the three immune conditions were
917 retained as the higher-level units rather than pooling all knockout queries as independent replicates.

918 For HIR-Bench, repeated random seeds generated from the same parameter configuration were not
919 treated as independent observations for prediction of the retrieval-flip regime. Cross-validation was
920 grouped by the label-determining (α , conflict) parameter pair, giving 28 effective parameter cells. Be-
921 cause each held-out parameter cell contains a single outcome class by construction, classifier perfor-
922 mance was summarized by pooled out-of-fold AUC rather than by averaging fold-specific AUC values.

923 For the patient-derived glioblastoma analysis, drug-pair splits from the same patient were treated as
924 dependent. Uncertainty in the recoverability gap was therefore estimated by cluster bootstrap resam-
925 pling at the patient level rather than at the drug-pair level, using 5,000 bootstrap resamples. Because
926 only four patients contributed evaluable drug-pair splits, these intervals were interpreted cautiously
927 and were supplemented by repeating the analysis after excluding the patient contributing the largest
928 number of comparisons. Recoverability differences were calculated within each drug-pair split before
929 aggregation, so the reported median gap is the median of paired supervised–unsupervised differences
930 rather than the difference between their marginal medians.

931 Tahoe-100M analyses were summarized across eligible drug–cell-line conditions or within-cell-line
932 drug pairs, according to the statistic being measured. Cell-level observations were used to estimate state-
933 specific responses and recoverability but were not treated as independent replicates. The two cellular-
934 state definitions—released cell-cycle state and the independently control-derived k -means partition—
935 were analysed in parallel and were not selected according to which produced the more favourable result.

936 All tests are interpreted in conjunction with the corresponding effect size, sample structure and
937 replication level. No inference from an external functional association, a state-recoverability gap or a
938 decision-relevance proxy is interpreted as evidence of therapeutic efficacy.

939 Data availability

940 All datasets analysed in this study are publicly available.

941 SciPlex3, Frangieh Perturb-CITE-seq and ZhaoSims2021 were obtained in harmonized form
942 from scPerturb¹⁴, Zenodo record 13350497 ([doi:10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4). The exact
943 harmonized files used in the analyses were SrivatsanTrapnell2020_sciplex3.h5ad for SciPlex3,
944 FrangiehIzar2021_RNA.h5ad and FrangiehIzar2021_protein.h5ad for Frangieh Perturb-CITE-seq, and
945 ZhaoSims2021.h5ad for the patient-derived glioblastoma analysis. The primary SciPlex3 data are avail-
946 able from the Gene Expression Omnibus under accession GSE139944¹⁵; the original Frangieh and
947 ZhaoSims2021 datasets are described in the corresponding publications^{17,32}. The CD34+ haematopoietic
948 progenitor dataset is available from the Gene Expression Omnibus under accession GSE306429.

949 Tahoe-100M¹⁶ was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed
950 through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M). This study uses plate 3
951 only, together with the released obs_metadata and sample_metadata tables. Analyses use the released

952 preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-
953 transformed space. No reported result is derived from the remaining thirteen plates.

954 The external functional evaluation used GDSC2 fitted dose-response data from the Genomics of
955 Drug Sensitivity in Cancer, release 8.5 (GDSC2_fitted_dose_response_270ct23.xlsx), available from
956 <https://www.cancerrxgene.org>. The compound-matching table used to align the SciPlex3 and GDSC2
957 drug panels (drug_match_table.csv) is provided with the analysis code.

958 Processed analysis inputs and the scripts used to construct them from the source datasets are pro-
959 vided in the accompanying repository. Source-data tables underlying each figure panel are provided
960 with the corresponding analysis outputs.

961 **Code availability**

962 All code used for data processing, retrieval, evaluation and statistical analysis is available at [https:](https://github.com/Boom5426/PopRetrieve)
963 [//github.com/Boom5426/PopRetrieve](https://github.com/Boom5426/PopRetrieve). The repository includes scripts for reconstructing the processed
964 analysis inputs, running the retrieval and evaluation experiments, and regenerating the source-data
965 tables used for the figures. A numerical consistency check re-derives the reported summary statis-
966 tics from the persisted experiment outputs, and a pipeline driver re-executes the main analyses from
967 the processed inputs. The Tahoe-100M analyses are implemented as a self-contained workflow under
968 analysis/tahoe_pilot/. Separate scripts quantify state-specific differential responses, recoverability of
969 drug-response structure and the disjoint state-ordering proxy under both cellular-state definitions, with
970 the corresponding per-condition and per-pair output tables retained for reproducibility.

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973 **Author contributions**

974 **[[Author contributions, CRediT taxonomy: conceptualization, methodology, software, formal analy-**
975 **sis, investigation, data curation, writing (original draft), writing (review and editing), visualization,**
976 **supervision, funding acquisition.]]**

977 **Competing interests**

978 **[[Competing interests statement. If none: "The authors declare no competing interests."]]**

979 **Additional information**

980 Supplementary Information is available for this paper: Extended Data Figs. 1–7, Supplementary Notes
981 1–4 and Supplementary Tables 1–4.

982 Correspondence and requests for materials should be addressed to **[[Name]]**.

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